

1. General Description

The EM74HCS09 is a quad 2-input AND gate with open-drain outputs and Schmitt-trigger inputs. Schmitt trigger inputs transform slowly changing input signals into sharply defined jitter-free output signals.

2. Features and Benefits

- Wide supply voltage range from 2.0 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- Latch-up performance exceeds 250 mA
- Unlimited rise and fall times
- Complies with JEDEC standards:
 - JESD8C (2.7 V to 3.6 V)
 - JESD7A (2.0 V to 6.0 V)
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 2 exceeds 3500 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

EM74HCS09

Quad 2-input AND Schmitt trigger with open-drain outputs

3. Ordering Information

Table 1. Ordering information

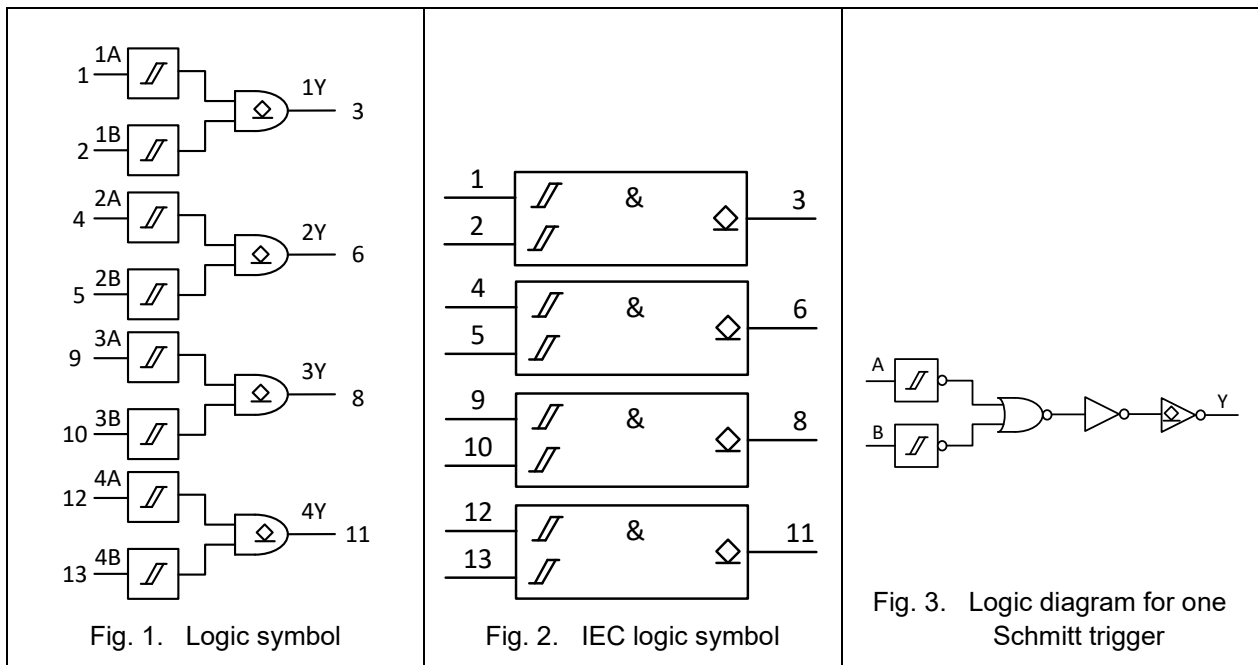
Type number	Topside marking	Package		
		Name	Description	Quantity
EM74HCS09D	HCS09□ XYYWW	SOP-14L	plastic small outline package; 14 leads; body width 3.9 mm	3000
EM74HCS09PW	HCS09□ XYYWW	TSSOP-14L	plastic thin shrink small outline package; 14 leads; body width 4.4 mm	3000

MARKING INFORMATION

NOTE: HCS09□ = Serial number and Internal code; XYYWW = Vendor Code and Date Code.

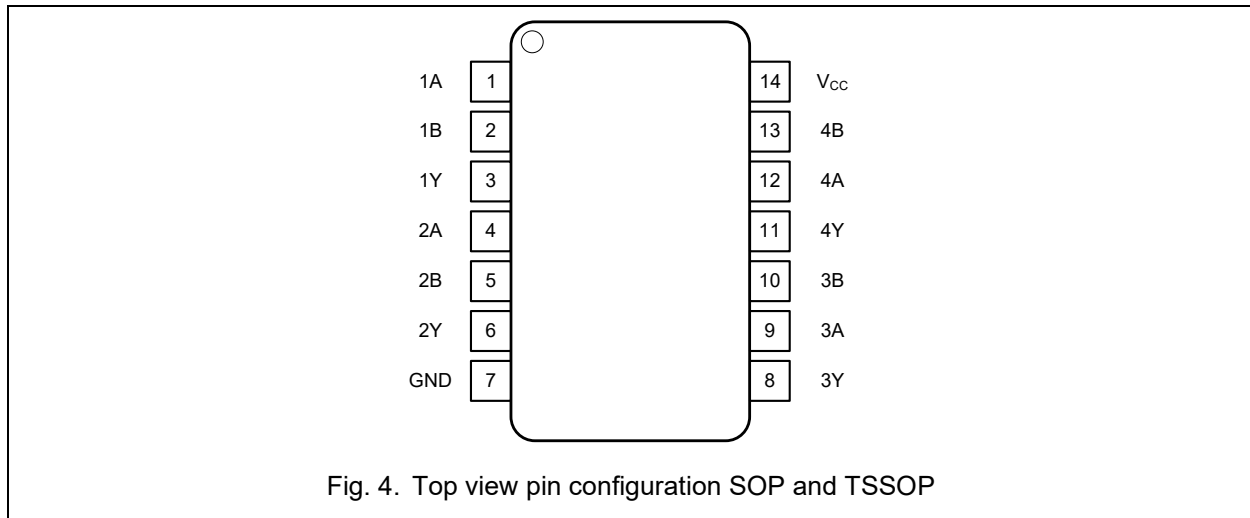


4. Function Diagram



5. Pinning Information

5.1. Pinning



5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
1A, 2A, 3A, 4A	1, 4, 9, 12	Data input
1B, 2B, 3B, 4B	2, 5, 10, 13	Data input
1Y, 2Y, 3Y, 4Y	3, 6, 8, 11	Data output
GND	7	Ground (0V)
V _{CC}	14	Supply voltage

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; Z:High impedance.

Control		Output
nA	nB	nY
L	L	L
L	H	L
H	L	L
H	H	Z

EM74HCS09

Quad 2-input AND Schmitt trigger with open-drain outputs

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	7.0	V
I_{IK}	input clamping current	$V_I < -0.5\text{ V}$ or $V_I > V_{CC} + 0.5\text{ V}$ [1]		± 20	mA
I_{OK}	output clamping current	$V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$ [1]		± 20	mA
V_O	output voltage		-0.5	$V_{CC} + 0.5\text{ V}$	V
I_O	output current	$V_O = -0.5\text{ V}$ to $(V_{CC} + 0.5\text{ V})$		± 35	mA
I_{CC}	supply current			70	mA
I_{GND}	ground current		-70		mA
P_{tot}	total power dissipation			500	mW
T_{stg}	storage temperature		-65	150	°C

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	EM74HCS09			Unit
			Min	Typ	Max	
V_{CC}	supply voltage		2.0	5.0	6.0	V
V_I	input voltage		0		V_{CC}	V
V_O	output voltage		0		V_{CC}	V
T_{amb}	ambient temperature		-40		125	°C

9.Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V). Typical values measured at $T_{amb} = 25^{\circ}\text{C}$ (unless otherwise noted).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
V_{OL}	LOW-level output voltage	$V_I = V_{T+}$ or V_{T-}						
		$I_O = 20\ \mu\text{A}; V_{CC} = 2.0\ \text{V}$			0.1		0.1	V
		$I_O = 20\ \mu\text{A}; V_{CC} = 4.5\ \text{V}$			0.1		0.1	V
		$I_O = 20\ \mu\text{A}; V_{CC} = 6.0\ \text{V}$			0.1		0.1	V
		$I_O = 6.0\ \text{mA}; V_{CC} = 4.5\ \text{V}$			0.33		0.4	V
		$I_O = 7.8\ \text{mA}; V_{CC} = 6.0\ \text{V}$			0.33		0.4	V
I_I	input leakage current	$V_I = V_{CC}$ or GND ; $V_{CC} = 6.0\ \text{V}$			± 1		± 1	μA
I_{OZ}	OFF-state output current	$V_I = V_{T+}$ or $V_{T-}; V_O = V_{CC}$ or GND; $V_{CC} = 6.0\ \text{V}$			± 5		± 10	μA
I_{CC}	supply current	$V_I = V_{CC}$ or GND ; $I_O = 0\ \text{A}$; $V_{CC} = 6.0\ \text{V}$			1		2	μA
C_i	input capacitance			7				pF

10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 6. Typical values measured at $T_{amb} = 25^{\circ}\text{C}$ (unless otherwise noted).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
t_{pd}	propagation delay	nA, nB to nY; see Fig. 5 [1]						
		$V_{CC} = 2.0\text{ V}$			35		40	ns
		$V_{CC} = 4.5\text{ V}$			20		25	ns
		$V_{CC} = 6.0\text{ V}$			15		20	ns
t_t	transition time	see Fig. 5 [2]						
		$V_{CC} = 2.0\text{ V}$			9		11	ns
		$V_{CC} = 4.5\text{ V}$			6		8	ns
		$V_{CC} = 6.0\text{ V}$			4		5	ns
C_{PD}	power dissipation capacitance	$C_L = 15\text{ pF}$; $f = 1\text{ MHz}$; $V_I = \text{GND to } V_{CC}$ [3]		6				pF

[1] t_{pd} is the same as t_{pZL} and t_{pLZ} .

[2] t_t is the same as t_{THL} and t_{TLH} .

[3] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

N = number of inputs switching;

$\Sigma(C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

10.1. Waveforms and test circuit

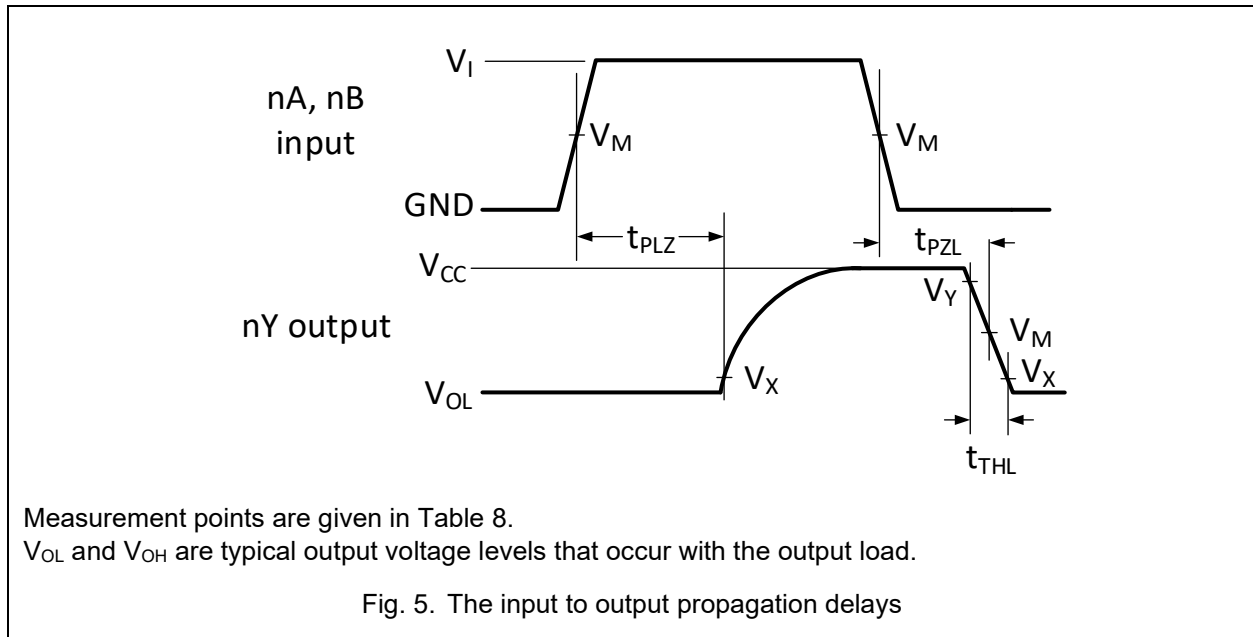
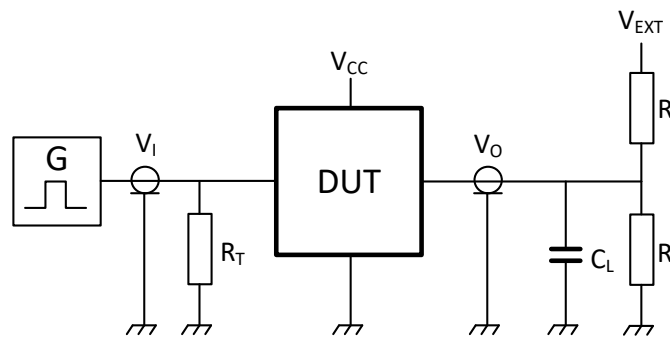
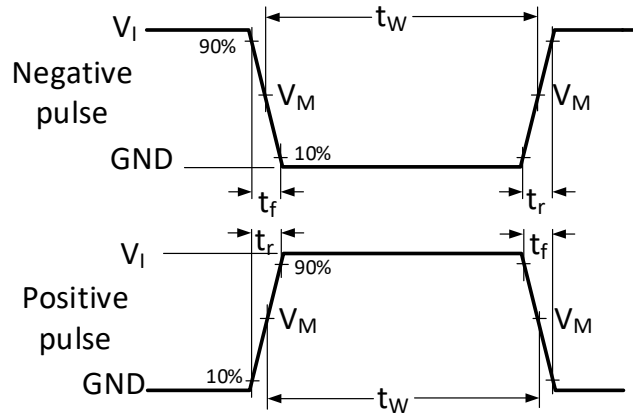


Table 8. Measurement points

Type	Input	Output		
	V_M	V_M	V_X	V_Y
EM74HCS09	$0.5V_{CC}$	$0.5V_{CC}$	$0.1V_{CC}$	$0.9V_{CC}$

EM74HCS09

Quad 2-input AND Schmitt trigger with open-drain outputs



Test data is given in Table 9.

Definitions for test circuit:

R_L = Load resistance.

C_L = Load capacitance including jig and probe capacitance.

R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator.

V_{EXT} = External voltage for measuring switching times.

Fig. 6. Test circuit for measuring switching times

Table 9. Test data

Type	Input		Load		V_{EXT}
	V_I	$t_r = t_f$	C_L	R_L	t_{PLH}, t_{PHL}
EM74HCS09	V_{CC}	≤ 2.5 ns	15 pF	500 Ω	open

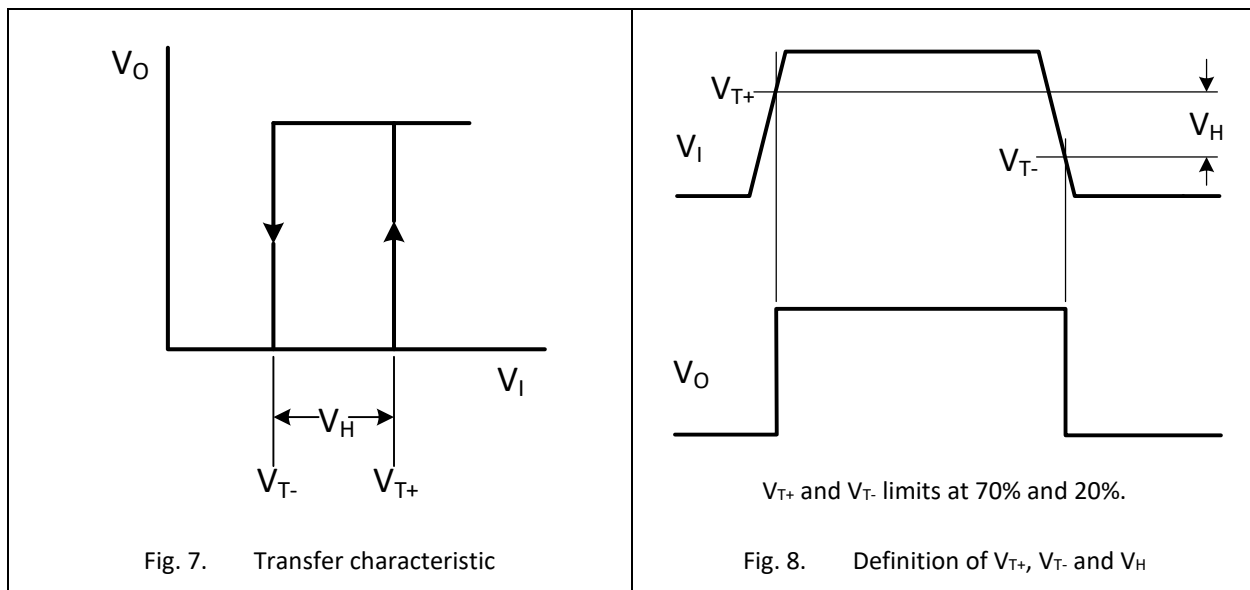
11. Transfer Characteristics

Table 10. Transfer characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V). Typical values are measured at $T_{amb} = 25^{\circ}\text{C}$.

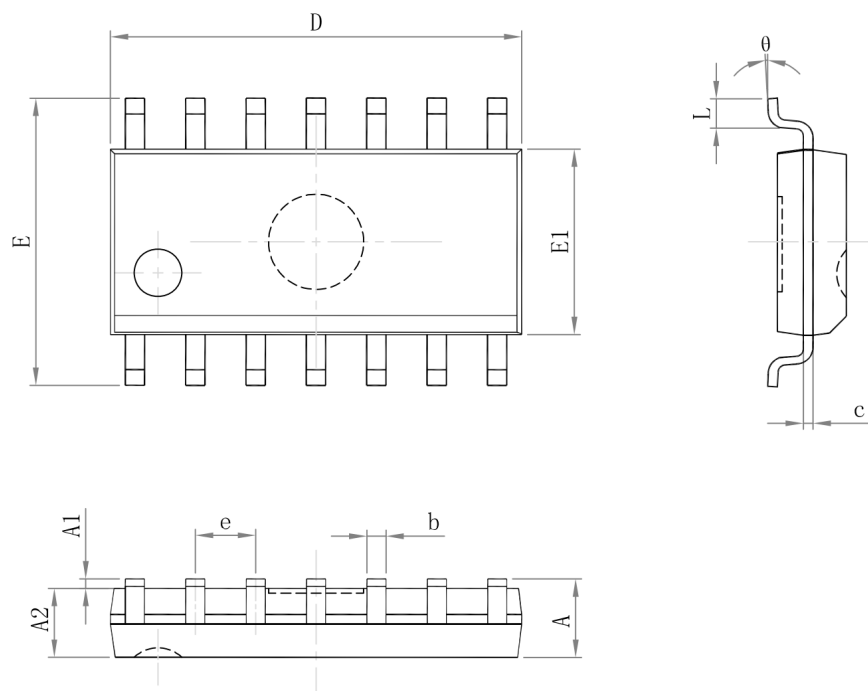
Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
V_{T+}	positive-going threshold voltage	see Fig. 7 and Fig. 8						
		$V_{CC} = 2.0\text{ V}$	0.7	1.21	1.5	0.7	1.5	V
		$V_{CC} = 4.5\text{ V}$	1.7	2.39	3.15	1.7	3.15	V
		$V_{CC} = 6.0\text{ V}$	2.1	3.05	4.2	2.1	4.2	V
V_{T-}	negative-going threshold voltage	see Fig. 7 and Fig. 8						
		$V_{CC} = 2.0\text{ V}$	0.3	0.6	0.9	0.3	0.9	V
		$V_{CC} = 4.5\text{ V}$	0.9	1.41	2.0	0.9	2.0	V
		$V_{CC} = 6.0\text{ V}$	1.2	1.9	2.6	1.2	2.6	V
V_H	hysteresis voltage	see Fig. 7 and Fig. 8						
		$V_{CC} = 2.0\text{ V}$	0.2	0.61	1.0	0.2	1.0	V
		$V_{CC} = 4.5\text{ V}$	0.4	0.97	1.4	0.4	1.4	V
		$V_{CC} = 6.0\text{ V}$	0.6	1.16	1.6	0.6	1.6	V

11.1. Waveforms transfer characteristics



12. Package Outline

SOP-14L

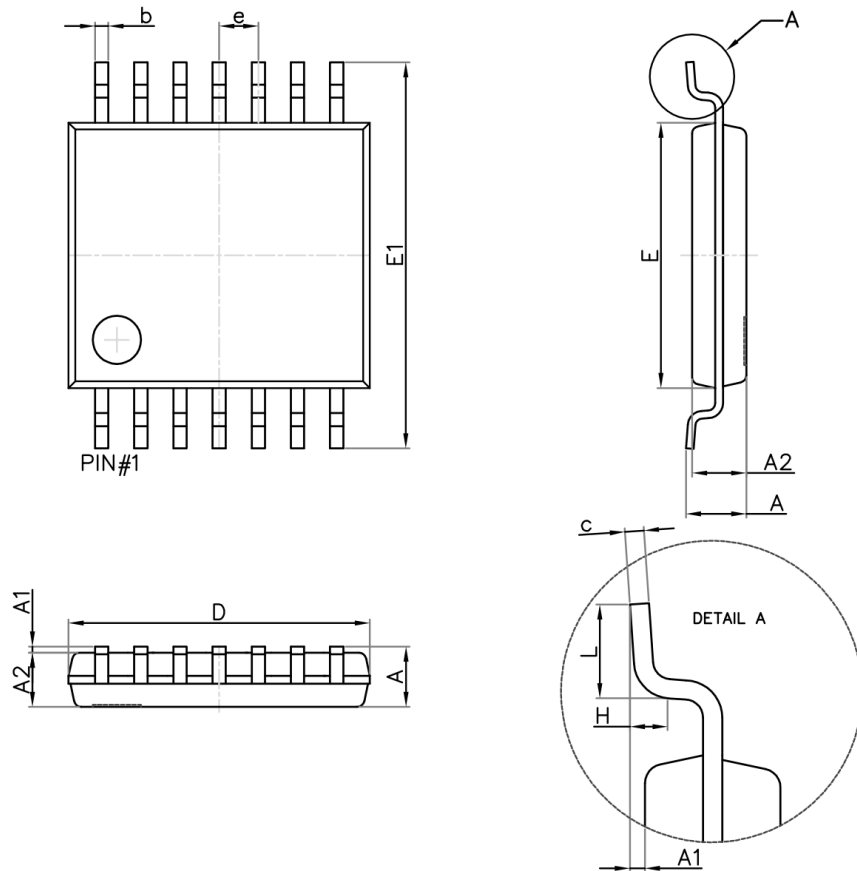


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	—	1.750	—	0.069
A1	0.100	0.250	0.004	0.010
A2	1.250	—	0.049	—
b	0.310	0.510	0.012	0.020
c	0.100	0.250	0.004	0.010
D	8.450	8.850	0.333	0.348
E	5.800	6.200	0.228	0.244
E1	3.800	4.000	0.150	0.157
e	1.270(BSC)		0.050(BSC)	
L	0.400	1.270	0.016	0.050
θ	0°	8°	0°	8°

EM74HCS09

Quad 2-input AND Schmitt trigger with open-drain outputs

TSSOP-14L



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
D	4.900	5.100	0.193	0.201
E	4.300	4.500	0.169	0.177
b	0.190	0.300	0.007	0.012
c	0.090	0.200	0.004	0.008
E1	6.250	6.550	0.246	0.258
A	—	1.200	—	0.047
A2	0.800	1.000	0.031	0.039
A1	0.050	0.150	0.002	0.006
e	0.65 (BSC)		0.026 (BSC)	
L	0.500	0.700	0.020	0.028
H	0.25(TYP)		0.01(TYP)	
θ	1°	7°	1°	7°

13. Tape and Reel Information

13.1. Carrier tape dimensions

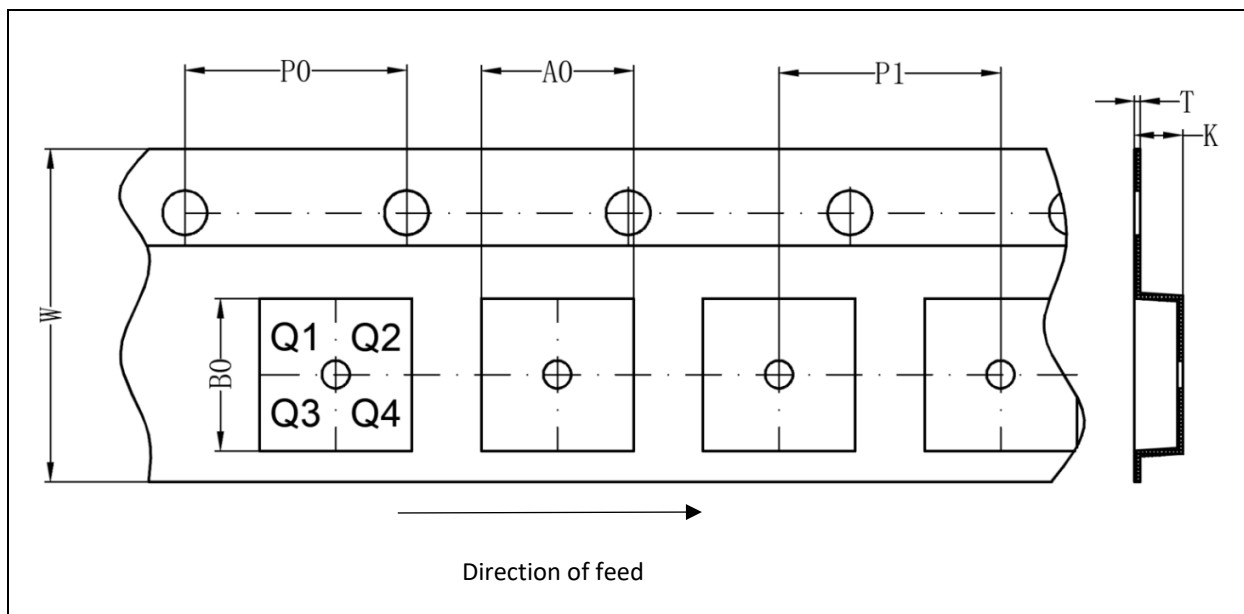


Table 11. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOP-14L	6.35	9.1	1.9	0.22	8	16	4	Q1
TSSOP-14L	6.7	5.45	1.6	0.25	8	12	4	Q1

EM74HCS09

Quad 2-input AND Schmitt trigger with open-drain outputs

13.2. Reel and box dimensions

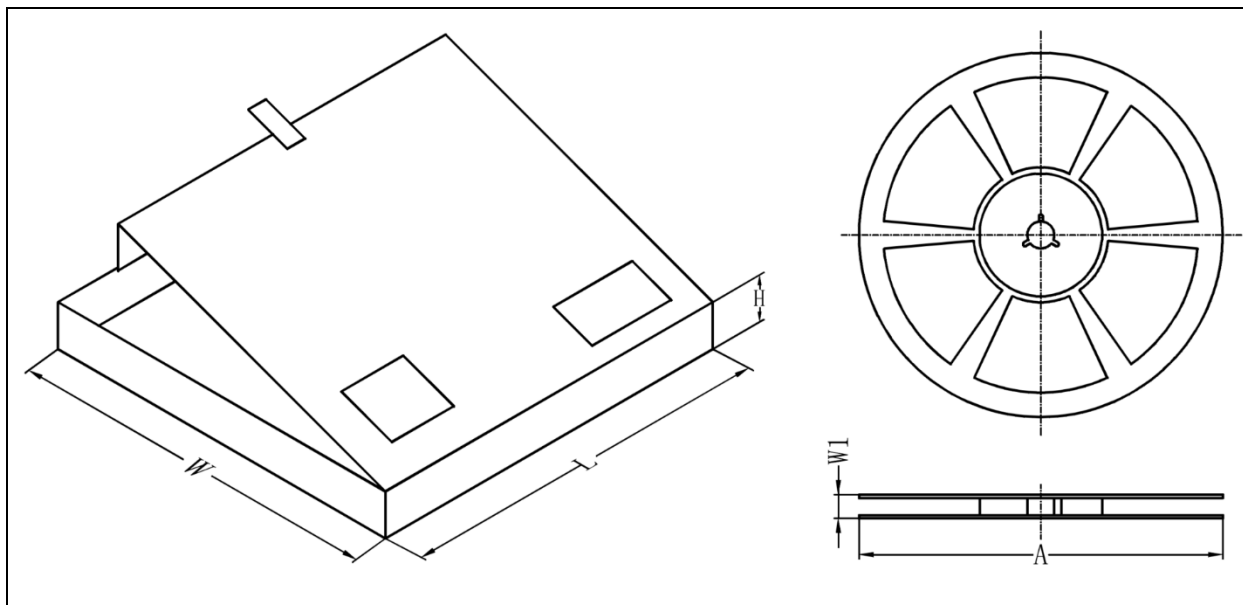


Table 12. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	SPQ (pcs) [1]	Reels per box	Outer box dimensions L×W×H(mm) [2]
SOP-14L	D	330	22.4	3000	1	358x340x50
TSSOP-14L	PW	330	18.4	3000	1	358x340x50

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

14. Abbreviations

Table 13. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model

15. Revision History

Table 14. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74HCS09 Rev. 1.1	Dec 15, 2025	Product datasheet		EM74HCS09 Rev. 1.0
Modifications:	- Table 1: topside marking information updated. - Section 13 added Tape and Reel Information.			
EM74HCS09 Rev. 1.0	Aug 30, 2024	Product datasheet		